REPORT FILE

VITYARTHI FLIPPED COURSE PROJECT

AI-Based Fake Product Review Detection System

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INTRODUCTION

In today's digital world, online shopping platforms contain a large number of product reviews. However, many of these reviews are fake or misleading. Fake reviews can confuse customers and reduce trust in online platforms.

This project aims to develop an **AI-based system** that can automatically detect whether a product review is **real or fake** using Machine Learning and Natural Language Processing (NLP).

2. Objective

The main objectives of this project are:

- To identify fake product reviews
- To analyze review text using NLP techniques
- To apply Machine Learning for classification
- To provide accurate and fast predictions
- To improve user trust in online platforms

3. Methodology

The system works in the following steps:

Step 1: Data Collection

A dataset of product reviews is created with labels:

- 0 → Real Review
- 1 → Fake Review

Step 2: Data Preprocessing

- Convert text to lowercase
- Remove special characters
- Clean the review text

Step 3: Feature Extraction

- TF-IDF (Term Frequency–Inverse Document Frequency) is used
- Converts text into numerical form for ML model

Step 4: Machine Learning Model

- Logistic Regression is used for classification
- The model learns patterns from data

Step 5: Rule-Based Detection

Additional rules are applied:

- Very short reviews → Fake
- Too many "!" → Fake
- Overly positive words with 5 rating → Suspicious

Step 6: Final Prediction

- Combines ML result + rule-based logic
- Outputs: **REAL or FAKE**

4. Tools and Technologies Used

☐ Programming Language: Python

☐ Libraries:

- pandas
- scikit-learn
- re (Regular Expressions)

6. Results

- The system successfully detects fake reviews
- Combines ML + rules for better accuracy
- Provides quick and efficient predictions

7. Applications

E-commerce platforms (Amazon, Flipkart)

Fraud detection systems

Customer feedback analysis

Online review filtering

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8. Advantages

. Easy to use

. Fast prediction

. Real-world application

. Improves customer trust

9. Limitations

- Small dataset may reduce accuracy
- Simple NLP approach
- Cannot detect highly complex fake reviews

10. Future Scope

- Use Deep Learning (LSTM, BERT)
- Build a web application
- Detect fake reviewers (not just reviews)
- Improve accuracy with large datasets

conclusion

This project demonstrates how Artificial intelligence can be used to solve real-world problems like fake review detection .By combining machine learning and rule – based logic, the system effectively identifies fake reviews and helps users makes better decisions.

THANK YOU